

Experiment 12

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(A)

Design a trigger which whenever there is an insertion on student table then, the currently inserted or deleted row should be printed as it as on the output console window.

Solution:

```
CREATE OR REPLACE FUNCTION FN_STUDENT_AUDIT()
RETURNS TRIGGER
LANGUAGE PLPGSQL
AS
$$
BEGIN
    IF TG_OP = 'INSERT' THEN
        RAISE NOTICE 'INSERTED ROW -> ID: %, NAME: %, AGE: %, CLASS: %',
            NEW.ID, NEW.NAME, NEW.AGE, NEW.CLASS;
        RETURN NEW;

    ELSIF TG_OP = 'DELETE' THEN
        RAISE NOTICE 'DELETED ROW -> ID: %, NAME: %, AGE: %, CLASS: %',
            OLD.ID, OLD.NAME, OLD.AGE, OLD.CLASS;
        RETURN OLD;

    END IF;

    RETURN NULL;
END;
$$;

CREATE TRIGGER TRG_STUDENT_AUDIT
AFTER INSERT OR DELETE
ON STUDENT
FOR EACH ROW
EXECUTE FUNCTION FN_STUDENT_AUDIT();

INSERT INTO STUDENT(NAME, AGE, CLASS)
VALUES('JOHN', 17, 11);

DELETE FROM STUDENT
WHERE NAME='JACK';
```

(B)

Design a POSTGRESQL triggers that:

Whenever a new employee is inserted in tbl_employee, a record should be added to tbl_employee_audit like:

"Employee name <emp_name> has been added at <current_time>"

Whenever an employee is deleted from tbl_employee, a record should be added to tbl_employee_audit like:

"Employee name <emp_name> has been deleted at <current_time>"

The solution must use PostgreSQL triggers.

Solution:

```
CREATE OR REPLACE FUNCTION AUDIT_EMPLOYEE_CHANGES()
RETURNS TRIGGER
LANGUAGE PLPGSQL
AS
$$
BEGIN
    IF TG_OP = 'INSERT' THEN
        INSERT INTO TBL_EMPLOYEE_AUDIT(message)
        VALUES ('Employee name ' || NEW.EMP_NAME || ' has been added at ' || NOW());

        RETURN NEW;

    ELSIF TG_OP = 'DELETE' THEN
        INSERT INTO TBL_EMPLOYEE_AUDIT(message)
        VALUES ('Employee name ' || OLD.EMP_NAME || ' has been deleted at ' || NOW());

        RETURN OLD;

    END IF;

    RETURN NULL;
END;
$$;

CREATE TRIGGER TRG_EMPLOYEE_AUDIT
AFTER INSERT OR DELETE
ON TBL_EMPLOYEE
FOR EACH ROW
EXECUTE FUNCTION AUDIT_EMPLOYEE_CHANGES();

INSERT INTO TBL_EMPLOYEE(EMP_NAME, EMP_SALARY)
VALUES ('ALAN', 50000);

DELETE FROM TBL_EMPLOYEE
WHERE EMP_NAME = 'ALAN';

SELECT * FROM TBL_EMPLOYEE_AUDIT;
```